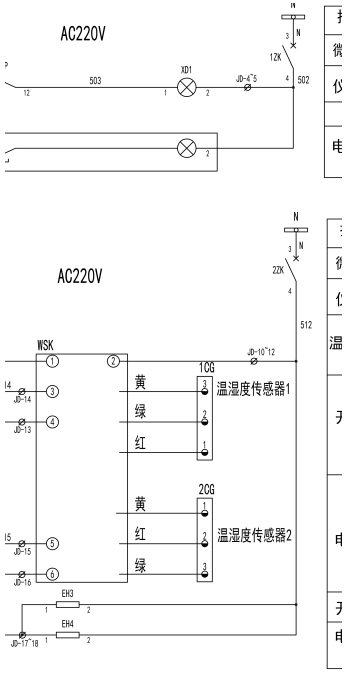


(c) Unknown contact state

Question

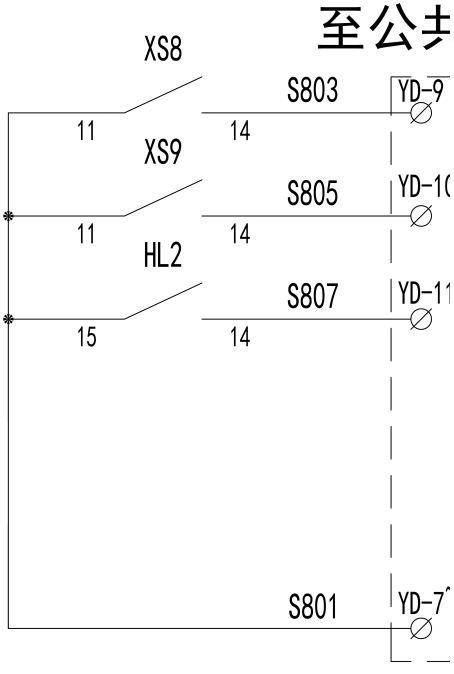
Is the connection between 2ZK:3 and EH4:2 fixed, or does it depend on 2ZK contact 3–4? Assume 1ZK contact 3–4 closed. Treat unlisted contacts as open.

Answer

It depends on that contact: connected when closed, disconnected when open.

Evidence summary

Closing 2ZK 3–4 gives 2ZK:3 → 2ZK:4 → EH4:2. Opening that contact disconnects the pair under the stated assumptions.



(d) Fault-hypothesis discrimination

Question

In an isolated continuity check from HL2:14, the readings are HL2:15 connected and XS8:14 connected. Exactly one of XS9 contact 14–11, XS8 contact 14–11 and HL2 contact 14–15 is stuck open; all other named contacts are closed and wires are intact. Which of these contacts could be stuck open? Treat unlisted contacts as open.

Answer

XS9 contact 14–11.

Evidence summary

Predicted readings (HL2:15, XS8:14): XS9 fault → (1,1); XS8 fault → (1,0); HL2 fault → (0,0). Only the XS9 fault matches both observations.